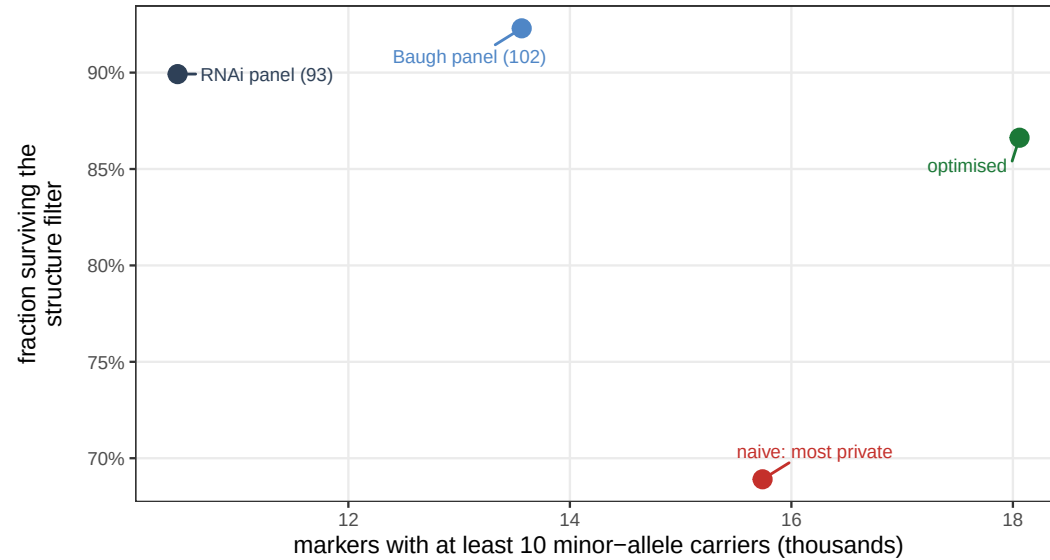


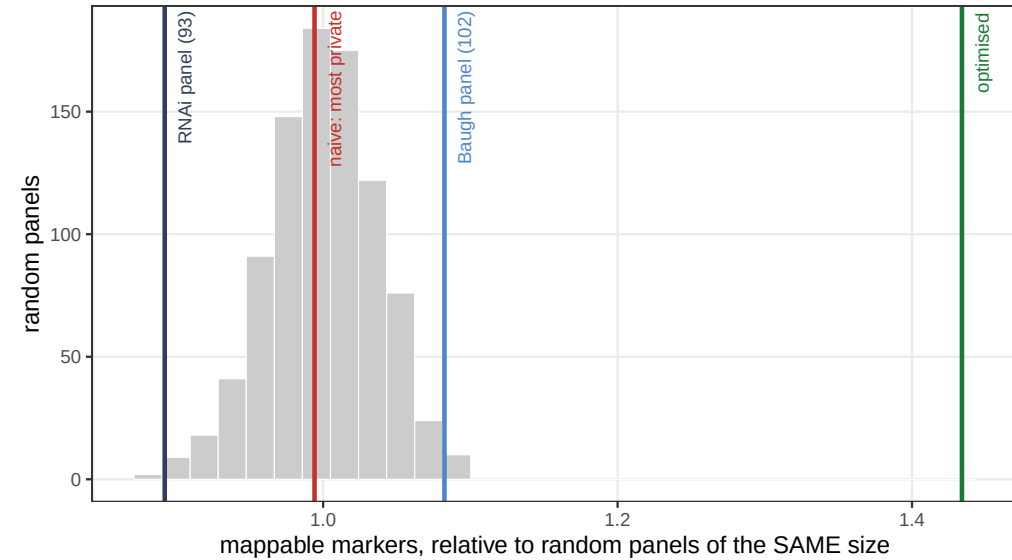
Where the naive panel loses

Selecting on private alleles finds nearly as much testable variation as the optimiser (15.7k against 17.8k markers) and then gives 31% of it back to population structure.



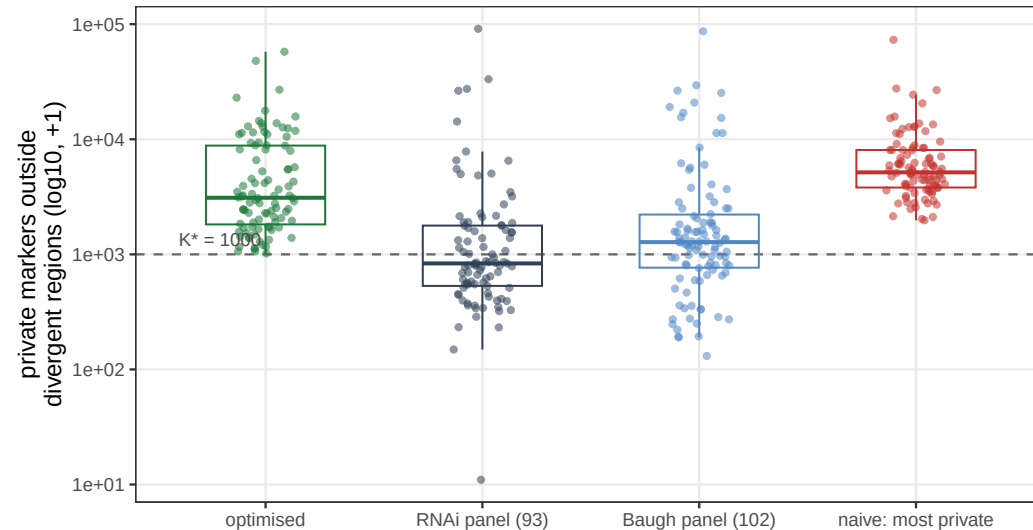
Each panel against 300 random panels of its own size

Panel size is divided out, so the three sizes share one axis. The naive panel sits on the null: ranking strains by private-allele count buys nothing at all for mapping. Note the optimised panel is not the least structured one -- its PC1 share is 0.137 against 0.076–0.083 for the two real panels. It tolerates structure wherever the markers survive the filter anyway.



The CONSTRAINT -- not the objective, and not what separates the top two

Dashed line is the identifiability floor. 54 of 93 strains in the RNAi panel and 40 of 102 in the Baugh panel fall below it; the lowest carries 10 private markers. The optimised and naive panels both clear it, and the NAIVE one clears it by more -- median 5,156 against 3,101, minimum 1,976 against 1,018. It wins this panel and still lands on the null in the one above. Identifiability and mappability are close to independent; panel A is where the two part company.



Simulated annealing over swaps

Each step drops one strain and adds another; proposals below the identifiability floor are rejected outright. The trace is STILL CLIMBING at the last iteration, so the reported panel is a lower bound on what this objective admits, not a converged optimum.

